Synergetic Meta-Learning: Unleashing the Power of Ensembled MAML Models for Adaptive Learning and Generalization

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Background and Motivation

Meta-learning, often referred as "learning to learn", is a category that aim to enable models to adapt to new tasks rapidly and with minimal data.

PROBLEM:
MAML has grown in prominence because to its capacity to learn a proper initialization that enables for quick adaptation to different jobs.

- MAML may not necessarily generalize well to a wide range of activities, particularly ones with large differences.
- MAML is known for its difficulty to converge during its training.

MOTIVATION:
The merging of different models, known as ensembling, has been shown to improve overall model performance, stability, and resilience. Ensembling within MAML training process may help in:

- Increase the generalization ability of MAML to diverse tasks
- Prevent the disruption of extreme data during the training process to decrease the number of training iterations and stable the convergence of the algorithm

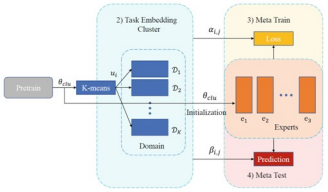
Literature Review

EEML: Ensemble Embedded Meta-learning (Li, Ren, & Wang, 2022)

- EEML employs gradient-based meta-learning, organizes tasks into clusters, and utilizes a weighted voting mechanism during meta-test.
- Advantages: Multi-expert cooperation enhances few-shot learning. Combines generic meta-learning with ensemble learning for improved results.
- Disadvantages: Hyper parameter tuning may impact effectiveness. Higher computation complexity as training several MAML learners in parallel

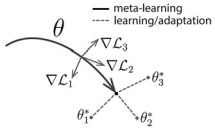
Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks (Finn, Abbeel, & Levine, 2017)

- Introduce a versatile meta-learning approach that quickly adapts diverse neural network architectures to novel tasks.
- Advantages: Adapting any gradient based methods. Fast adaptation with limited data. Generalization across different tasks.
- Disadvantages: Sensitivity to hyperparameters. Requires a modest amount of task-specific data. Complexity in implementation.



Original MAML Overview

Model-Agnostic Meta-Learning (MAML) aims to train a model in such a way that it can quickly adapt to new, unseen tasks with only a few examples, by learning a good initialization of its parameters.



Model-Agnostic Meta-Learning

Require: $p(T)$: distribution over tasks
Require: α, β: step size hyperparameters
randomly initialize θ
while not done **do**
Sample batch of tasks $T_i \sim p(T)$
for all T_i **do**
Evaluate $\nabla_\theta L_{T_i}(f_\theta)$ with respect to K examples
Compute adapted parameters with gradient descent: $\theta_0' = \theta - \alpha \nabla_\theta L_{T_i}(f_\theta)$
end for
Update $\theta \leftarrow \theta - \beta \nabla_\theta \sum_{T_i \sim p(T)} L_{T_i}(f_{\theta_0'})$
end while

Research Approach

Methodology:
In this research we explore how ensembling techniques can be utilized when it comes to Meta-Learning/Few-Shot Learning. The implementation for this is quite unique in nature as compared with the traditional machine learning models as MAML involves local as well as global parameters.

Approach:

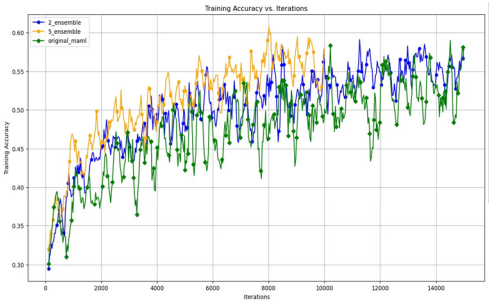
- Initialize the global parameters for MAML
- For each batch of task, initialize N models with identical local parameters. Within each task these local parameters are averaged out so that it does not overfit a particular task.
- These local parameters are then used to update the global parameters. This helps MAML generalize better to new unseen tasks.
- We have also tried experimenting with different model architectures including changing the base neural network and learning rates.

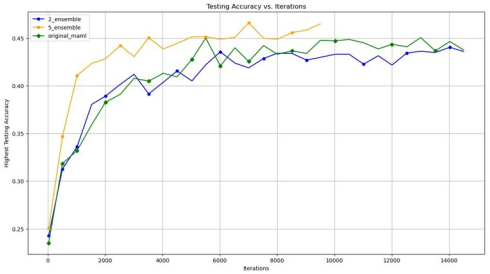
Meta Learner Level Model-Agnostic Meta-Learning Algorithm

Initialize global parameters θ.
for each iteration **do**
Sample a batch of tasks $\mathcal{T}_i$.
for each task $T_i \in \mathcal{T}_i$ **do**
Initialize N models with identical parameters $\theta_{i,1}, \theta_{i,2}, \ldots, \theta_{i,N}$.
for each model $m \in \{1, 2, \ldots, N\}$ **do**
for i training iterations **do**
Evaluate $\nabla_{\theta_{i,m}} \mathcal{L}_{T_i}(f_{\theta_{i,m}})$ with respect to K examples.
Update $\theta_{i,m} \leftarrow \theta_{i,m} - \alpha \nabla_{\theta_{i,m}} \mathcal{L}_{T_i}(f_{\theta_{i,m}})$.
end for
Evaluate $\mathcal{L}_{i,m}(f_{\theta_{i,m}})$ with respect to query set examples.
Evaluate $\mathcal{A}_{i,m}(f_{\theta_{i,m}})$ as the query set accuracy of task i ensemble m model.
end for
Evaluate $\mathcal{L}_{T_i} \leftarrow \sum_{m \in \mathcal{N}} \frac{\mathcal{A}_{i,m}(f_{\theta_{i,m}})}{\sum_{m' \in \mathcal{N}} \mathcal{A}_{i,m'}(f_{\theta_{i,m'}})} \mathcal{L}_{i,m}(f_{\theta_{i,m}})$.
end for
Update $\theta \leftarrow \theta - \nabla_\theta \beta \sum_{T_i \in \mathcal{T}_i} \mathcal{L}_{T_i}$.
end for

Experimental Results

- Dataset: MiniImageNet
- Split: 80 classes in training set, 20 classes in testing set
- Problem setup: 5-way 1-shot in support set, 15 images in query set, 5 gradient updates in training, 10 gradient updates in testing
- Preprocessing: Image Normalization
- Batch Loading: 4 tasks each batch/iteration, 2500 batches/iterations for 1 epoch





Model	Highest Training Accuracy	Highest Testing Accuracy
2 ensemble	0.667	0.44
5 ensemble	0.657	0.466
original maml	0.677	0.45

Conclusion and Future Work

Conclusions:

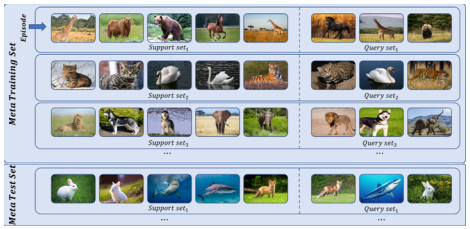
- Ensembling within MAML increased the performance for around 1%
- Ensembling within MAML stabilized the convergence process, helping the algorithm increase the accuracy in earlier iterations
- Ensembling within MAML costs much more computational resources since multiple models are trained for each task

Future Work:

- Examine the algorithm with different settings: different number of ways/shots, different number of ensembles, different number of tasks for each iteration, different dataset
- Implement different ensembling methods to replace the simple weighted average idea
- Try the local ensembling idea on other types of meta-learners

Terminology

- **N-way**: Number of classes to classify
- **K-shot**: Number of samples for each one of the N classes
- **Support Set**: The N-way K-shot "training set" for the learner to learn in the training process
- **Query Set**: The "testing set" in training to reflect the learner's learning



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